

Application Build and Release Automation Lab

Assignment 2 – Practice Examination 2

Student Name: Ayush Bhargav

Roll Number: 241501877

Q1 – Linux Project Setup — Photo 1

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~  
$ # QUESTION 1- Linux Project Setup  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~  
$ mkdir Ayush-GLA-HealthSystem  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~  
$ cd Ayush-GLA-HealthSystem  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ mkdir src docs config backup  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ touch health_info.txt  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ nano health_info.txt  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ pwd  
/c/Users/Ayush Bhargav/Ayush-GLA-HealthSystem  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ cd src  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem/src  
$ cd ..  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ ls  
backup/  config/  docs/  health_info.txt  src/  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ cat health_info.txt  
Student Name: Ayush Bhargav  
Roll Number: 241501877  
Subject Name: Application Build and Release Automation Lab  
Project Name: Campus Health Appointment System  
Semester: 5th  
  
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem  
$ find .  
.  
./backup  
./config  
./docs  
./health_info.txt  
./src
```

Q2 – Git Version Control — Photo 1

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem
$ git init
Initialized empty Git repository in C:/Users/Ayush Bhargav/Ayush-GLA-HealthSystem/.git/

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$
git config --global user.name "Ayush Bhargav"

git config --global user.email "ayushbhargav008@gmail.com"

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    health_info.txt

nothing added to commit but untracked files present (use "git add" to track)

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add .
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Initial commit for health appointment system"
[master (root-commit) e12428d] Initial commit for health appointment system
 1 file changed, 5 insertions(+)
 create mode 100644 health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git diff
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/health_info.txt b/health_info.txt
index f92cd3f..a12cb85 100644
--- a/health_info.txt
+++ b/health_info.txt
@@ -3,3 +3,7 @@ Roll Number: 241501877
 Subject Name: Application Build and Release Automation Lab
 Project Name: Campus Health Appointment System
 Semester: 5th
+
+
+Project Objective: To develop a Campus Health Appointment System for managing health appointments.
+Student Program/Course: B.Tech Computer Science and Engineering
```

Q2 – Git Version Control — Photo 2

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Add project objective and course details"
[master 9adf74e] Add project objective and course details
1 file changed, 4 insertions(+)

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline
9adf74e (HEAD -> master) Add project objective and course details
e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean
```

Q3 – Independent Feature Development — Photo 1

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch appointment-feature

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git switch appointment-feature
Switched to branch 'appointment-feature'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git branch --show-current
appointment-feature

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ touch src/appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ nano src/appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git add src/appointment.txt
warning: in the working copy of 'src/appointment.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git commit -m "Add appointment module"
[appointment-feature 5d7e071] Add appointment module
1 file changed, 15 insertions(+)
create mode 100644 src/appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git branch
* appointment-feature
  master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git branch --show-current
appointment-feature

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (appointment-feature)
$ git switch master
Switched to branch 'master'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch --show-current
master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log appointment-feature --oneline
```

Q3 – Independent Feature Development — Photo 2

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch --show-current
master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log appointment-feature --oneline
5d7e071 (appointment-feature) Add appointment module
9adf74e (HEAD -> master) Add project objective and course details
e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
```

Q4 – Merge and Resolve Conflict — Photo 1

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git switch master
Already on 'master'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git merge appointment-feature
Updating 9adf74e..5d7e071
Fast-forward
 src/appointment.txt | 15 ++++++
 1 file changed, 15 insertions(+)
 create mode 100644 src/appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ ls src
appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline --graph --all
* 5d7e071 (HEAD -> master, appointment-feature) Add appointment module
* 9adf74e Add project objective and course details
* e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ part a complete
bash: part: command not found

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git switch system-update
Switched to branch 'system-update'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ git branch --show-current
system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ nano health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ git add health_info.txt
warning: in the working copy of 'health_info.txt', LF will be replaced by CRLF the next time Git touches it

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ git commit -m "Update project name in system-update"
[system-update 4fbeb9c] Update project name in system-update
 1 file changed, 4 insertions(+)
```

Q4 – Merge and Resolve Conflict — Photo 2

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (system-update)
$ git switch master
Switched to branch 'master'

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ nano health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git add health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git commit -m "Update project name in master"
[master cfbde31] Update project name in master
1 file changed, 3 insertions(+)

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git merge system-update
Auto-merging health_info.txt
CONFLICT (content): Merge conflict in health_info.txt
Automatic merge failed; fix conflicts and then commit the result.

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ nano health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ grep -n "<<<<<<|=====|>>>>>>" health_info.txt
12:<<<<<< HEAD
14:=====
17:>>>>>> system-update
19:<<<<<< HEAD
21:=====
23:>>>>>> system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git add health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git commit -m "Resolve health info merge conflict"
[master b105d9c] Resolve health info merge conflict
```


Q4 – Merge and Resolve Conflict — Photo 3

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ nano health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ grep -n "<<<<<<<\|=====\|>>>>" health_info.txt
12:<<<<<<< HEAD
14:=====
17:>>>>>>> system-update
19:<<<<<<< HEAD
21:=====
23:>>>>>>> system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git add health_info.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master|MERGING)
$ git commit -m "Resolve health info merge conflict"
[master b105d9c] Resolve health info merge conflict

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline --graph --all
*   b105d9c (HEAD -> master) Resolve health info merge conflict
| \
|  * 4fbeb9c (system-update) Update project name in system-update
|  * cfbde31 Update project name in master
| /
* 5d7e071 (appointment-feature) Add appointment module
* 9adf74e Add project objective and course details
* e12428d Initial commit for health appointment system
```


Q5 – Repository Demonstration — Photo 1

```
Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ find .
.
./.git
./.git/COMMIT_EDITMSG
./.git/config
./.git/description
./.git/HEAD
./.git/hooks
./.git/hooks/applypatch-msg.sample
./.git/hooks/commit-msg.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/post-update.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/pre-commit.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-push.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-receive.sample
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/update.sample
./.git/index
./.git/info
./.git/info/exclude
./.git/logs
./.git/logs/HEAD
./.git/logs/refs
./.git/logs/refs/heads
./.git/logs/refs/heads/appointment-feature
./.git/logs/refs/heads/master
./.git/logs/refs/heads/system-update
./.git/objects
./.git/objects/07
./.git/objects/07/851b1f481725074ec4403bb2e428b581bd07eb
./.git/objects/13
./.git/objects/13/df8c5142772cc05f8e0f538be180bed7ee5a18
./.git/objects/3c
./.git/objects/3c/70add63ee8279c8cd722882b055d345513664e
./.git/objects/4f
./.git/objects/4f/beb9c90ae3ab1c14ab777ef819a039ab63e71c
./.git/objects/55
./.git/objects/55/6dc77603483d19eeb6eea37927230d0c32a732
./.git/objects/5d
./.git/objects/5d/7e071824eedb9d48cdfdc388815304350ba5fe
./.git/objects/67
./.git/objects/67/97b317ce04c548cade407546155c42c4ba7fa0
./.git/objects/72
./.git/objects/72/239288bbe31994d229229cd8f6754f5214ab6f
./.git/objects/72/a999f1650820776b6e1bec3dd34d4bf8a99c58
./.git/objects/9a
```

Q5 – Repository Demonstration — Photo 2

```
./git/objects/f1
./git/objects/f1/a8a45e7ca10e02e36d11972ee374fbff463ea2
./git/objects/f9
./git/objects/f9/2cd3fb1d08620ad03c5b731930f1e1364511d6
./git/objects/fc
./git/objects/fc/6381096af595064dfa1d6cdb6fca2d97322286
./git/objects/info
./git/objects/pack
./git/ORIG_HEAD
./git/refs
./git/refs/heads
./git/refs/heads/appointment-feature
./git/refs/heads/master
./git/refs/heads/system-update
./git/refs/tags
./backup
./config
./docs
./health_info.txt
./src
./src/appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git status
On branch master
nothing to commit, working tree clean

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline --graph --all
*    b105d9c (HEAD -> master) Resolve health info merge conflict
| \
|  * 4fbeb9c (system-update) Update project name in system-update
|  * | cfbde31 Update project name in master
| /
* 5d7e071 (appointment-feature) Add appointment module
* 9adf74e Add project objective and course details
* e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master
  system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ ls src
appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ cat health_info.txt
Student Name: Ayush Bhargav
Roll Number: 241501877
```

Q5 – Repository Demonstration — Photo 3

```
* e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git branch
  appointment-feature
* master
  system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ ls src
appointment.txt

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ cat health_info.txt
Student Name: Ayush Bhargav
Roll Number: 241501877
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th

Project Objective: To develop a Campus Health Appointment System for managing health appointments.
Student Program/Course: B.Tech Computer Science and Engineering

<<<<<<< HEAD
Project Name: Campus Health Appointment System - Main Update
=====

Project Name: Campus Health Appointment System - System Update
>>>>>> system-update

<<<<<<< HEAD
Project Name: Campus Health Appointment System - Main Update
=====
Project Name: Campus Health Appointment System - System Update
>>>>>> system-update

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ git log --oneline --graph --all
*   b105d9c (HEAD -> master) Resolve health info merge conflict
| \
| * 4fbeb9c (system-update) Update project name in system-update
* | cfbde31 Update project name in master
| /
* 5d7e071 (appointment-feature) Add appointment module
* 9adf74e Add project objective and course details
* e12428d Initial commit for health appointment system

Ayush Bhargav@LAPTOP-PK65VJLN MINGW64 ~/Ayush-GLA-HealthSystem (master)
$ |
```